Even function: A function $f(x)$ is said to be even if $f(-x) = f(x)$

odd function: A function $f(x)$ is said to be odd if $f(-x) = -f(x)$.

(a) $f(x) = x + |x-1|$

putting $x = 3$, we get,

$f(3) = 3 + |3-1|$

$= 5$

putting $x = -3$ we get,

$f(-3) = -3 + |-3-1|$

$= -3 + 4$

(b) $f(x) = \begin{cases} 5+x & \text{when } x<3 \\ 5 & \text{when } x=3 \\ 5+x & x>3 \end{cases}$

when $x<3$,

$f(x) = 5+x$

$\therefore f(-2) = 5+(-2) = 5-2 = 3$

when $x=3$, $f(x) = 5$

$\therefore f(3) = 5$

when $x>3$, $f(x) = 5-x$.

$\therefore f(4) = 5-4$

$= 1$

27) $f(x) = x+3$

$f(x)$ is exist for all real values of x.

So the domain of $f(x)$ is $D_f = \mathbb{R}$.

let, $y = f(x) = x+3$

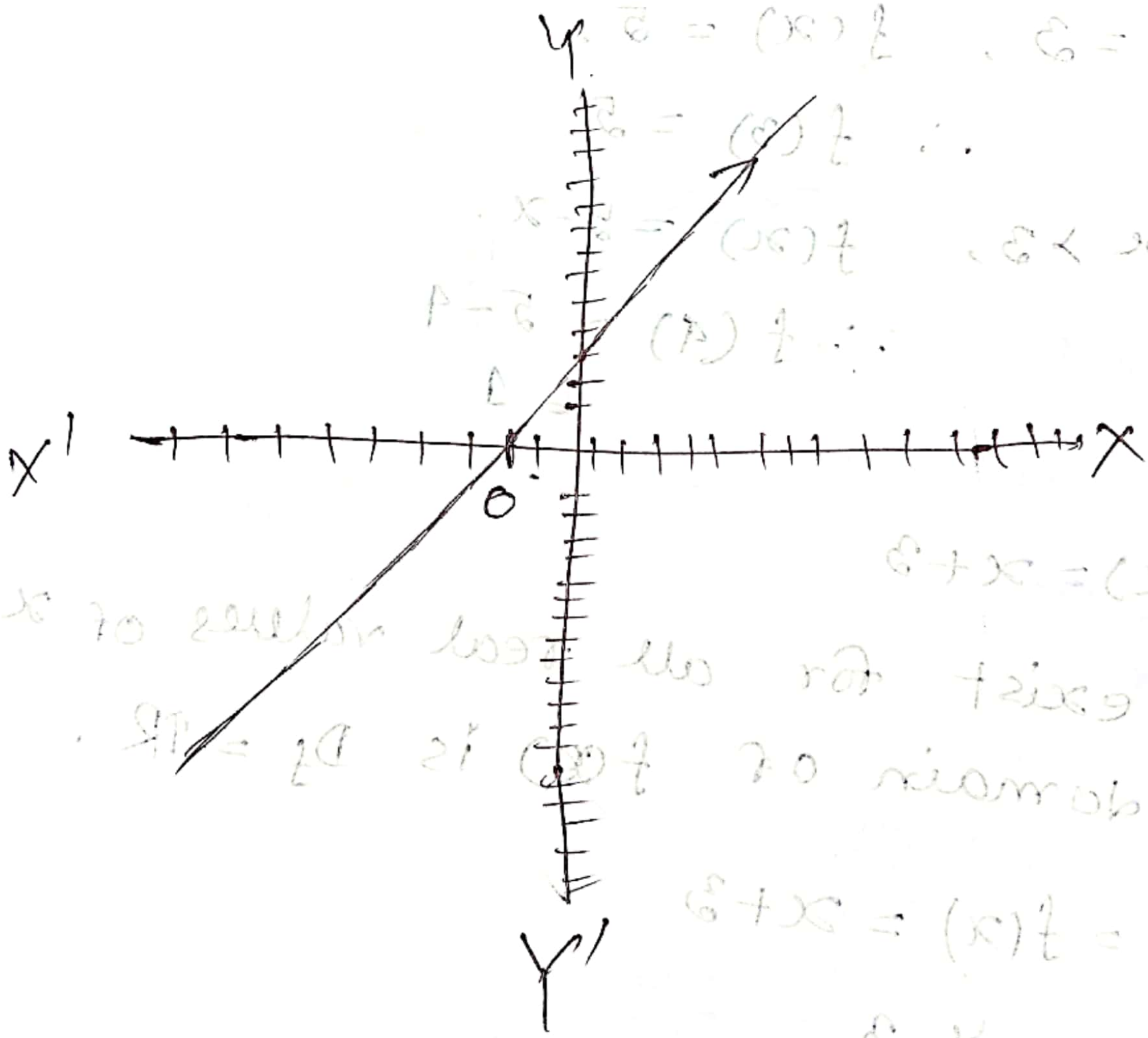
$\therefore x = y-3$

x is exist for all real values of x is exist in its domain for all real values of y.

So the range is $R_f = \mathbb{R}$

again, $y = f(x) = x+3$

x	-10	-9	-5	-3	0	1	2	3
y	-7	-6	-2	0	3	4	5	6



2b| $f(x) = \sqrt{x-1}$

$f(x)$ is exist if $x \geq 1$.

So the domain of $f(x)$ is $D_f = [1, \infty)$

let, $y = f(x) = \sqrt{x-1}$

$y^2 = x-1$.

$\therefore x = y^2 + 1$

x is exist in its domain in its domain for all real values of y.

So the range of $f(x)$ is $R_f = \mathbb{R}$.

2a| $f(x) = \dfrac{x^2-4}{x-2}$

$f(x)$ is ~~exi~~ define for all real values of x except $x = 2$.

So the domain of the given function is

$D_f = \mathbb{R} - \{2\}$

let, $y = f(x) = \dfrac{x^2-4}{x-2} = x+2$

$\cancel{xy - y = x^2 - 4}$ $\quad x = y - 2$

$\cancel{x^2 - xy = 4 - y}$

$\cancel{x(}$

x is exist in its domain for all real values of y except $y = 4$.

So the range of the given function is $R_f = \mathbb{R} - \{4\}$

(Qu-88) important,

~~example~~-

Qu| $f(x) = \begin{cases} -1 & \text{when } x < 0 \\ 0 & \text{when } x = 0 \\ 1 & \text{"} \quad x > 0 \end{cases}$

Domain of the given function, ~~D_f (-∞, 0)~~

$D_f = (-\infty, 0) \cup \{0\} \cup (0, \infty)$

$= (-\infty, \infty)$

$= \mathbb{R}$

for $x < 0$, $f(x) = -1$

for $x = 0$, $f(x) = 0$

for $x > 0$, $f(x) = 1$

So, the range of the given function is, $R_f = \{-1, 0, 1\}$

9)

$$f(x) = \begin{cases} x & \text{when } 0 \le x < \frac{1}{2} \\ 1 & \text{when } x = \frac{1}{2} \\ 1-x & \text{when } \frac{1}{2} < x < 1 \end{cases}$$

Domain of the given function is

$D_f = [0, \frac{1}{2}) \cup \{\frac{1}{2}\} \cup (\frac{1}{2}, 1)$

$= [0, 1)$

For $0 \le x < \frac{1}{2}$, $0 \le f(x) < \frac{1}{2}$

For $x = \frac{1}{2}$, $f(x) = 1$

For $\frac{1}{2} < x < 1$, $0 < f(x) < \frac{1}{2}$

So the range of $f(x)$ is,

$R_f = [0, \frac{1}{2}) \cup \{1\} \cup [0, \frac{1}{2})$

$= [0, \frac{1}{2}) \cup \{1\}$

Q7|

$$f(x) = \begin{cases} 3 & \text{when } -3 \le x < -1 \\ 3x-3 & \text{when } 0 < x \le 1 \\ -6x-3 & \text{when } -1 \le x \le 0 \end{cases}$$

Domain of the given function is,

$D_f = [-3, -1) \cup (0, 1] \cup [-1, 0]$

$= [-3, 1]$

for $-3 \le x < -1$, $f(x) = 3$

for $0 < x \le 1$, $-3 < f(x) \le 0$

for $-1 \le x \le 0$, $-3 \le f(x) \le 3$

$\therefore$ So the range of the function is

~~$R_f = [-3, 0] \cup [$~~

$R_f = \{3\} \cup (-3, 0] \cup [-3, 3]$

$= [-3, 3]$

৩২।
$$f(x) = \begin{cases} 5, & \text{when } 0 < x < 1 \\ 10, & \text{when } 1 < x \le 2 \\ 15, & \text{when } 2 < x \le 3 \end{cases}$$

Domain of the function is,

$D_f = (0, 1) \cup (1, 2] \cup (2, 3]$

$= (0, 3]$

for, $0 < x < 1$, $f(x) = 5$.

for, $1 < x \le 2$, $f(x) = 10$

for, $2 < x \le 3$, $f(x) = 15$

So, the range of the function is,

$R_f = \{5\} \cup \{10\} \cup \{15\}$

$= \{5, 10, 15\}$

chapter-3

$f(x) = |x|$

when, $x > 0$, $f(x) = |x|$
$= x$

$$\lim_{x \to 0^+} f(x) = \lim_{h \to 0} f(0+h)$$

$$= \lim_{h \to 0} f(0+h)$$

$$= \lim_{h \to 0} (0+h)$$

$$= 0$$

when $x < 0$, $f(x) = |x|$
$= -x$

$$\lim_{x \to 0^-} f(x) = \lim_{h \to 0} f(0-h)$$

$$= \lim_{h \to 0} (-0+h)$$

$$= 0$$

when $x = 0$, $f(x) = |x|$

$$\therefore f(0) = |0|$$
$$= 0$$

$$\therefore \lim_{x \to 0^+} f(x) = \lim_{x \to 0^-} f(x) = f(0)$$

Thus $f(x)$ is continuous at $x = 0$.

(9) $f(x) = \begin{cases} x & \text{when } x \geq 0 \\ -x & \text{when } x < 0 \end{cases}$

when $x > 0$, $f(x) = x$

$$\lim_{x \to 0^+} f(x) = \lim_{h \to 0} f(0+h)$$

$$= \lim_{h \to 0} (0+h)$$

$$= 0$$

when $x < 0$, $f(x) = -x$

$$\lim_{x \to 0^-} f(x) = \lim_{h \to 0} f(0-h)$$

$$= \lim_{h \to 0} \{-(0-h)\}$$

$$= \lim_{h \to 0} h$$

$$= 0$$

when $x = 0$, $f(x) = x$

$\therefore f(0) = 0$

Thus $f(x)$ is continuous at $x = 0$.

Q) $f(x) = \begin{cases} \frac{|x|}{x} & \text{when } x \neq 0 \\ \frac{x}{1} & \text{when } x = 0 \end{cases}$

when $x > 0$, $f(x) = \frac{x}{x} = 1$.

$\lim_{x \to 0^+} f(x) = \lim_{h \to 0} f(0+h) = \lim_{h \to 0} f(h)$

$= \lim_{h \to 0} 1 = 1$

when $x < 0$, $f(x) = \frac{-x}{x} = -1$.

$\lim_{x \to 0^-} f(x) = \lim_{h \to 0} f(0-h) = \lim_{h \to 0} f(-h)$

$= \lim_{h \to 0} -1 = -1$

when $x = 0$,

$f(x) = 1$

$\therefore f(0) = 1$

$\lim_{x \to 0^+} f(x) \neq \lim_{x \to 0^-} f(x) \neq f(0)$

Thus $f(x)$ is not continous at $x = 0$

221 : $f(x) = \begin{cases} x, & 0 \le x < \frac{1}{2} \\ 1-x, & \frac{1}{2} \le x \le 1 \end{cases}$

when $x > \frac{1}{2}$, $f(x) = 1-x$.

$$\lim_{x \to \frac{1}{2}^+} f(x) = \lim_{h \to 0} f\left(\frac{1}{2}+h\right)$$

$$= \lim_{h \to 0} \left(1 - \frac{1}{2} - h\right)$$

$$= \lim_{h \to 0} \frac{1}{2} - h$$

$$= \frac{1}{2}$$

when $x < \frac{1}{2}$, $f(x) = x$

$$\lim_{x \to \frac{1}{2}^-} f(x) = \lim_{h \to 0} f\left(\frac{1}{2} - h\right)$$

$$= \lim_{h \to 0} \frac{1}{2} - h$$

$$\lim_{h \to 0} \frac{1}{2} - h$$

$$= \frac{1}{2}$$

when $x = \frac{1}{2}$, $f(x) = 1-x$.

$$f\left(\frac{1}{2}\right) = 1 - \frac{1}{2}$$

$$= \frac{1}{2}$$

Thus $f(x)$ is continuous at $x = \frac{1}{2}$

$$R.f'\left(\frac{1}{2}\right) = \lim_{h \to 0} \frac{f\left(\frac{1}{2}+h\right) - f\left(\frac{1}{2}\right)}{h}$$

$$= \lim_{h \to 0} \frac{\frac{1}{2} - h - \frac{1}{2}}{h}$$

$$= \lim_{h \to 0} \frac{1-2h-1}{2h}$$

$$= \lim_{h \to 0} -1$$

$$= -1$$

$$L.f'\left(\frac{1}{2}\right) = \lim_{h \to 0} \frac{f\left(\frac{1}{2}-h\right) - f\left(\frac{1}{2}\right)}{-h}$$

$$= \lim_{h \to 0} \frac{\frac{1}{2} - h - \frac{1}{2}}{-h}$$

$$= \lim_{h \to 0} \frac{1-2h-1}{-2h}$$

$$= \lim_{h \to 0} 1 = 1.$$

$\therefore Rf'\left(\frac{1}{2}\right) \neq L.f'\left(\frac{1}{2}\right)$

Thus. $f(x)$ is not differentiable at $x = \frac{1}{2}$

২৪। $f(x) = \begin{cases} 1 & , -\infty < x < 0 \\ 1+\sin x & , 0 \le x \le \frac{\pi}{2} \\ 2+(x-\frac{\pi}{2})^2 & , \frac{\pi}{2} \le x \le \infty \end{cases}$

At $x = \frac{\pi}{2}$,

when $x \geq \frac{\pi}{2}$, $f(x) = 2 + (x - \frac{\pi}{2})^2$

$$\lim_{x \to \frac{\pi}{2}^+} f(x) = \lim_{h \to 0} f\left(\frac{\pi}{2} + h\right)$$

$$= \lim_{h \to 0} \left\{ 2 + \left(\frac{\pi}{2} + h - \frac{\pi}{2}\right)^2 \right\}$$

$$= \lim_{h \to 0} (2 + h^2)$$

$$= 2$$

when $x < \frac{\pi}{2}$, $f(x) = 1 + \sin x$

$$\lim_{x \to \frac{\pi}{2}^-} f(x) = \lim_{h \to 0} f\left(\frac{\pi}{2} - h\right)$$

$$= \lim_{h \to 0} \left\{ 1 + \sin\left(\frac{\pi}{2} - h\right) \right\}$$

$$= \lim_{h \to 0} (1 + \cos h)$$

$$= 2$$

when $x = \frac{\pi}{2}$, $f(x) = 1 + \sin x$,

$$\therefore f\left(\frac{\pi}{2}\right) = 1 + \sin\frac{\pi}{2}$$

$$= 2$$

$$\therefore \lim_{x\to\frac{\pi}{2}^+} f(x) = \lim_{x\to\frac{\pi}{2}^-} f(x) = f\left(\frac{\pi}{2}\right)$$

Thus $f(x)$ is continuous at $x = \frac{\pi}{2}$

$$R.f'\left(\frac{\pi}{2}\right) = \lim_{h\to 0} \frac{f\left(\frac{\pi}{2}+h\right) - f\left(\frac{\pi}{2}\right)}{h}$$

$$= \lim_{h\to 0} \frac{2+h^2-2}{h}$$

$$= \lim_{h\to 0} h$$

$$= 0$$

$$L.f'\left(\frac{\pi}{2}\right) = \lim_{h\to 0} \frac{f\left(\frac{\pi}{2}-h\right) - f\left(\frac{\pi}{2}\right)}{-h}$$

$$= \lim_{h\to 0} \frac{1+\cos h-2}{-h}$$

$$= \lim_{h\to 0} \frac{1-\cos h}{h}$$

$$= \lim_{h\to 0} \frac{2\sin^2 h/2}{h}$$

$$= \lim_{h\to 0} \frac{\sin h/2}{h/2} \times \lim_{h\to 0} \sin h/2$$

$$= 1 \times 0$$

$\therefore R.f'\left(\frac{\pi}{2}\right) = L.f'\left(\frac{\pi}{2}\right)$

Hence, $f(x)$ is differentiable at $x = \frac{\pi}{2}$.

At $x = 0$,

when $x > 0$, $f(x) = 1 + \sin x$

$$\lim_{x \to 0^+} f(x) = \lim_{h \to 0} f(0+h)$$

$$= \lim_{h \to 0} \{1 + \sin(0+h)\}$$

$$= \lim_{h \to 0} (1 + \sin h)$$

$$= 1$$

when $x < 0$, $f(x) = 1$

$$\lim_{x \to 0^-} f(x) = \lim_{h \to 0} f(0-h)$$

$$= \lim_{h \to 0} 1$$

$$= 1$$

when $x = 0$, $f(x) = 1 + \sin x$

$\therefore f(0) = 1 + \sin 0$

$= 1$

$$\therefore \lim_{x \to 0^+} f(x) = \lim_{x \to 0^-} f(x) = f(0)$$

Thus $f(x)$ is continous at $x=0$.

$$R \cdot f'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{1+\sin h - 1}{h}$$

$$= \lim_{h \to 0} \frac{\sin h}{h}$$

$$= 1$$

$$L \cdot f'(0) = \lim_{h \to 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \to 0} \frac{1-1}{-h}$$

$$= 0$$

$$\therefore \quad R \cdot f'(0) \neq L \cdot f'(0)$$

Hence $f(x)$ is not differentiable at $x=0$.

2n) $f(x) = \begin{cases} 3+2x, & -3/2 \le x \le 0 \\ 3-2x, & 0 \le x \le 3/2 \\ -3-2x, & x \ge 3/2 \end{cases}$

when. $x > \frac{3}{2}$, $f(x) = -3-2x$

$\therefore \lim_{x \to \frac{3}{2}^+} f(x) = \lim_{h \to 0} f\left(\frac{3}{2}+h\right)$

$= \lim_{h \to 0} \left\{-3-2\left(\frac{3}{2}+h\right)\right\}$

$= \lim_{h \to 0} (-6-2h)$

$= -6$

when $x < \frac{3}{2}$, $f(x) = 3-2x$

$\therefore \lim_{x \to \frac{3}{2}^-} f(x) = \lim_{h \to 0} f\left(\frac{3}{2}-h\right)$

$= \lim_{h \to 0} \left\{3-2\left(\frac{3}{2}-h\right)\right\}$

$= \lim_{h \to 0} 2h$

$= 0$

when $x = \frac{3}{2}$, $f(x) = 3-2x$

$f\left(\frac{3}{2}\right) = 3-2\cdot\frac{3}{2}$

$= 0$

$\therefore \lim_{x \to \frac{3}{2}^+} f(x) \neq \lim_{x \to \frac{3}{2}^-} = f\left(\frac{3}{2}\right)$

Thus $f(x)$ is not continuous at $x = \frac{3}{2}$

$$R.f'\left(\frac{3}{2}\right) = \lim_{h \to 0} \frac{f\left(\frac{3}{2}+h\right) - f\left(\frac{3}{2}\right)}{h}$$

$$= \lim_{h \to 0} \frac{-6-2h-0}{h}$$

$$= \lim_{h \to \infty} \frac{-6-2h}{h}$$

$$= -\infty$$

$$L.f'\left(\frac{3}{2}\right) = \lim_{h \to 0} \frac{f\left(\frac{3}{2}-h\right) - f\left(\frac{3}{2}\right)}{-h}$$

$$= \lim_{h \to 0} \frac{2h-0}{-h}$$

$$= \lim_{h \to 0} -2$$

$$= -2$$

$$\therefore R.f'\left(\frac{3}{2}\right) \neq L.f'\left(\frac{3}{2}\right)$$

Thus $f'(x)$ ~~$f(\frac{3}{2})$~~ is not differentiable at $x =$

Sol: $f(x) = \frac{x^2+3x-10}{x-2}$ for, $x \neq 2$

$$\lim_{x\to 2^+} f(x) = \lim_{h\to 0} f(2+h)$$

$$= \lim_{h\to 0} \frac{(2+h)^2+3(2+h)-10}{2+h-2}$$

$$= \lim_{h\to 0} \frac{4+4h+h^2+6+3h-10}{h}$$

$$= \lim_{h\to 0} \frac{h^2+7h}{h}$$

$$= \lim_{h\to 0} h+7$$

$$= 7$$

$$\lim_{x\to 2^-} f(x) = \lim_{h\to 0} f(2-h)$$

$$= \lim_{h\to 0} \frac{(2-h)^2+3(2-h)-10}{2-h-2}$$

$$= \lim_{h\to 0} \frac{4-4h+h^2+6-3h-10}{-h}$$

$$= \lim_{h\to 0} \frac{h^2-7h}{-h}$$

$$= \lim_{h \to 0} 7 - h$$

$$= 7$$

$$\therefore \lim_{x \to 2^+} f(x) = \lim_{x \to 2^-} f(x) = 7$$

The function will be continuous at $x = 2$, if $f(2) = 7$.